

AWS

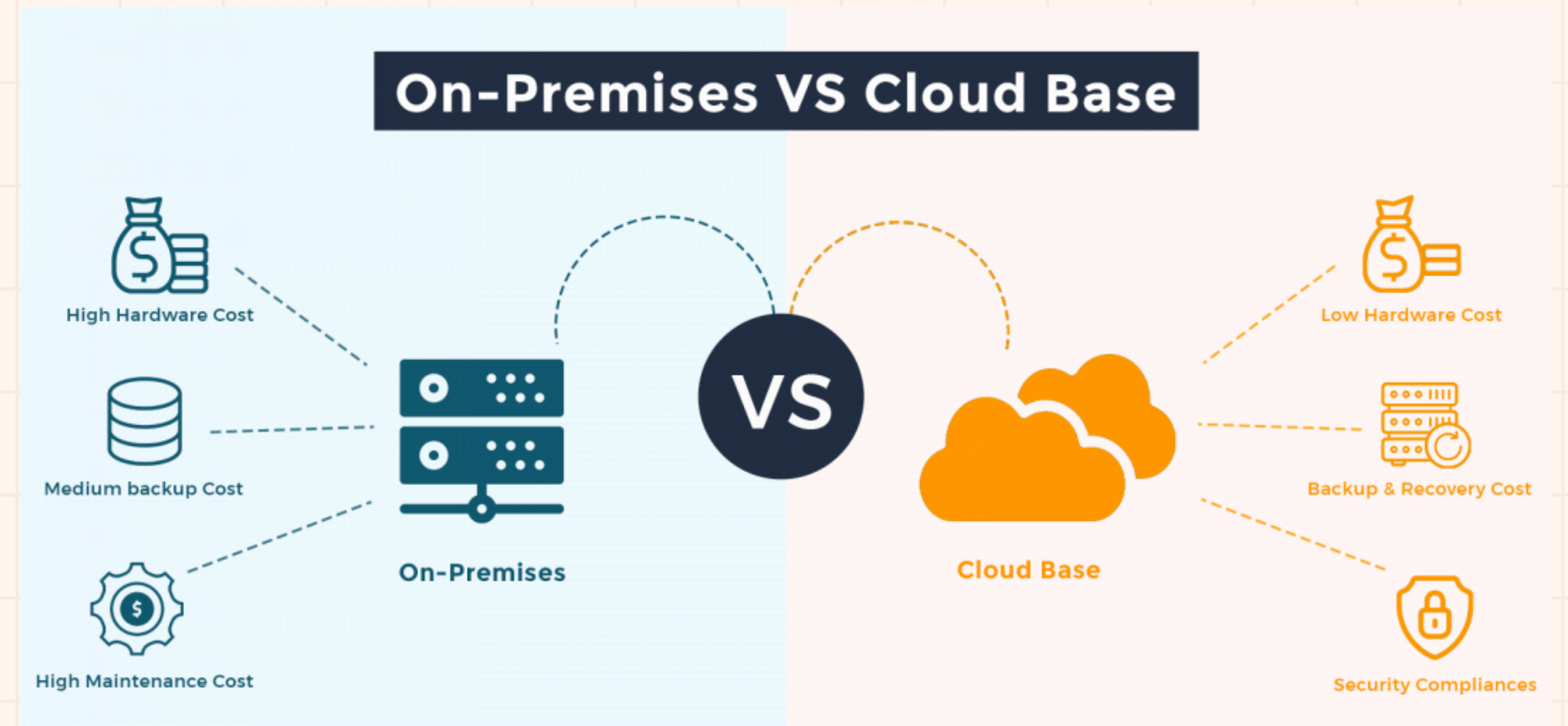
AMAZON WEB SERVICES



CLOUD COMPUTING FUNDAMENTALS AND ARCHITECTURE IN AWS

Context

Amazon Web Services (AWS) is a global cloud computing platform that provides on-demand access to computing power, storage, networking, and advanced digital services. It has fundamentally transformed how organizations build, deploy, and scale technological systems by replacing fixed infrastructure with elastic, pay-as-you-go resources.



Aplicaciones

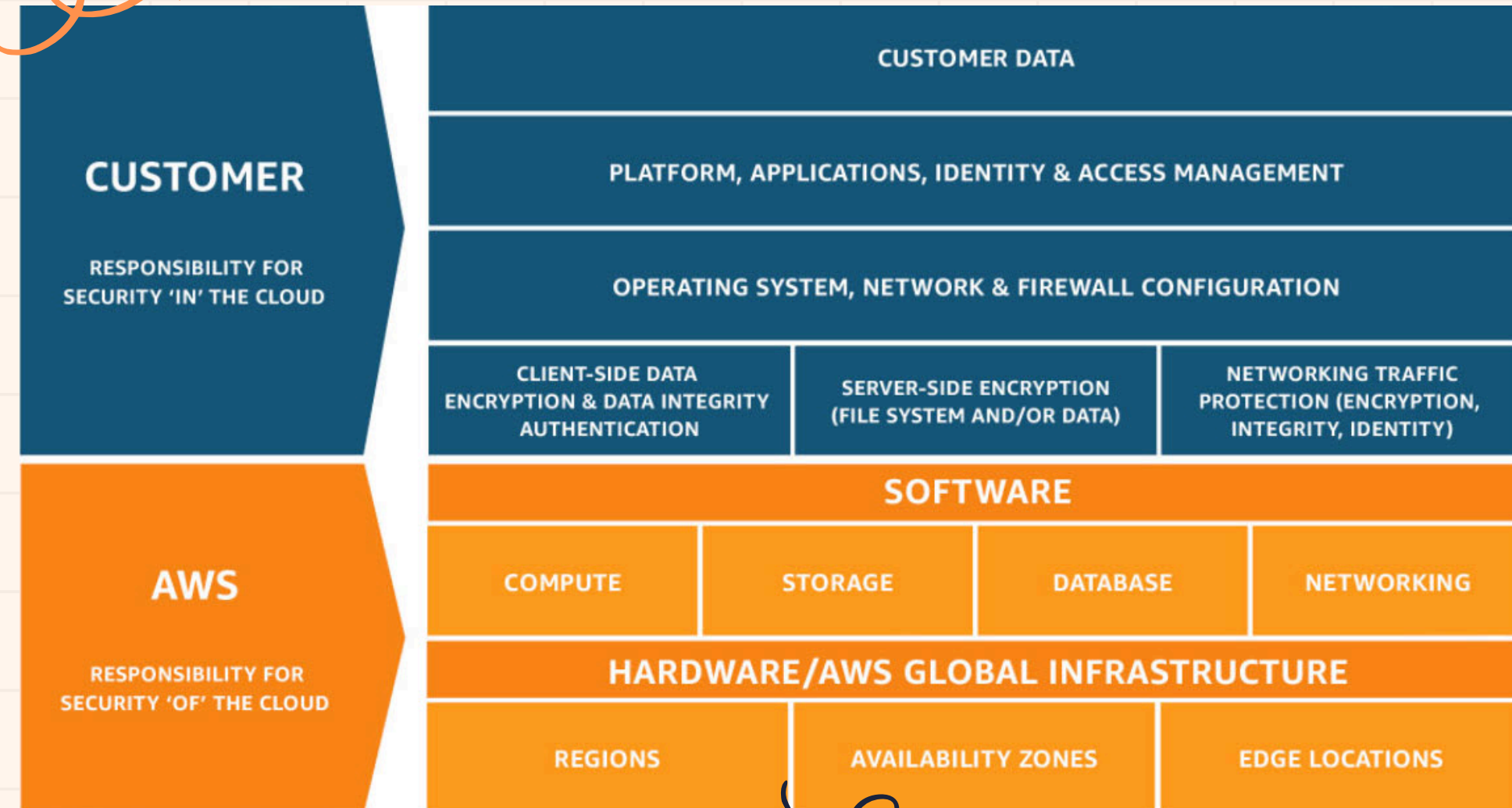
- Finance (risk modeling, data lakes)
- Healthcare (biomedical analytics)
- E-commerce (scalability)
- Scientific research (HPC)

AWS Global Infrastructure

- **Regions:** physical locations in the world. Importance: optimize latency, minimize costs, meet compliance laws.
- **Availability Zones (AZs):** isolated locations known as Availability Zones.
- **Edge Locations:** data centers to cache content closer to end-users to reduce latency.



Security OF the cloud



Security IN the cloud

CORE AWS SERVICES



COMPUTE

- **Amazon EC2** (Elastic Compute Cloud): virtual servers, choose the OS and the hardware power.
- **AWS Lambda**: Serverless compute. Upload code and by a trigger run. Without servers.
- **AWS Elastic Beanstalk**: For deploying web apps.



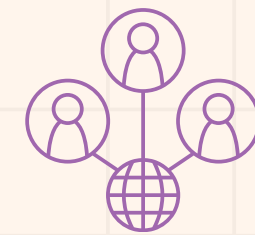
STORAGE

- **Amazon S3** (Simple Storage Service): object storage files.
- **Amazon EBS** (Elastic Block Store): Block storage, persistent storage for virtual servers.
- **Amazon Glaciar**: low-cost storage for data archiving and long-term backup.



DATABASE

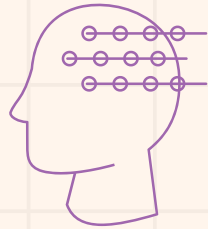
- **Amazon RDS** (Relational Database Service): Manage SQL databases.
- **Amazon DynamoDB**: managed NoSQL database.



NETWORKING

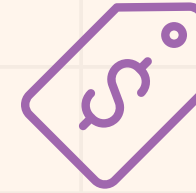
- **Amazon VPC** (Virtual Private Cloud): isolated section where you can launch resources in a virtual network.
- **Amazon CloudFront**: Content Delivery Network (CDN) for distribution of static web content.
- **Amazon Route 53**: cloud Domain Name System (DNS).

CORE AWS SERVICES



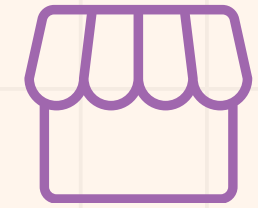
MACHINE LEARNING

- **Amazon SageMaker:** Build, train, and deploy models at scale.
- **Amazon Bedrock:** Build generative AI applications without managing infrastructure.
- **Amazon Rekognition:** Pre-trained computer vision service.
- **Amazon Comprehend:** Natural language processing service.
- **Amazon Forecast:** Time-series forecasting service powered by machine learning. No deep ML expertise required.
- **Amazon Textract:** Extract structured data from documents such as PDFs and scanned files.



COSTS

- **AWS Cost Explorer:** Visualize, analyze, and understand usage and spending patterns.
- **AWS Budgets:** Set cost thresholds and receive alerts when limits are exceeded.
- **AWS Cost and Usage Report (CUR):** Detailed billing data for advanced cost analysis.
- **AWS Savings Plans:** Commitment-based pricing model for reduced compute costs.



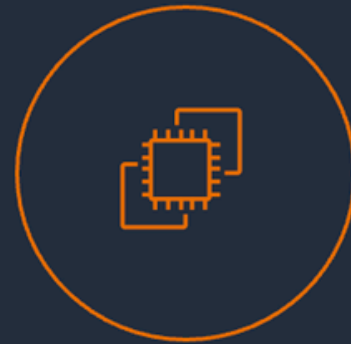
AWS MARKETPLACE

- **AWS Marketplace:** Digital catalog of third-party software that runs on AWS.
- **Pre-configured enterprise software:** Deploy solutions directly into your AWS environment.
- **Integrated billing:** All third-party subscriptions appear in your AWS bill.
- **Specialized solutions:** Security tools, ML models, financial software, and data products.

Groups of
services

AWS COMPUTE SERVICES

EC2



Amazon EC2 services

Amazon Elastic Compute
Cloud (Amazon EC2)
Amazon EC2 Auto Scaling
EC2 Image Builder
Amazon Lightsail

Containers



Container services

Amazon Elastic Container
Service (Amazon ECS)
Amazon ECS Anywhere
Amazon Elastic
Kubernetes Service
(Amazon EKS)
Amazon EKS Anywhere
Amazon Elastic Container
Registry (Amazon ECR)
AWS Batch

Serverless



Serverless compute

AWS Fargate
AWS Lambda

On-premises/Edge



On-premises and edge compute

AWS Local Zones
AWS Dedicated Local Zones
AWS Outposts
AWS Wavelength

Cost



Cost optimization

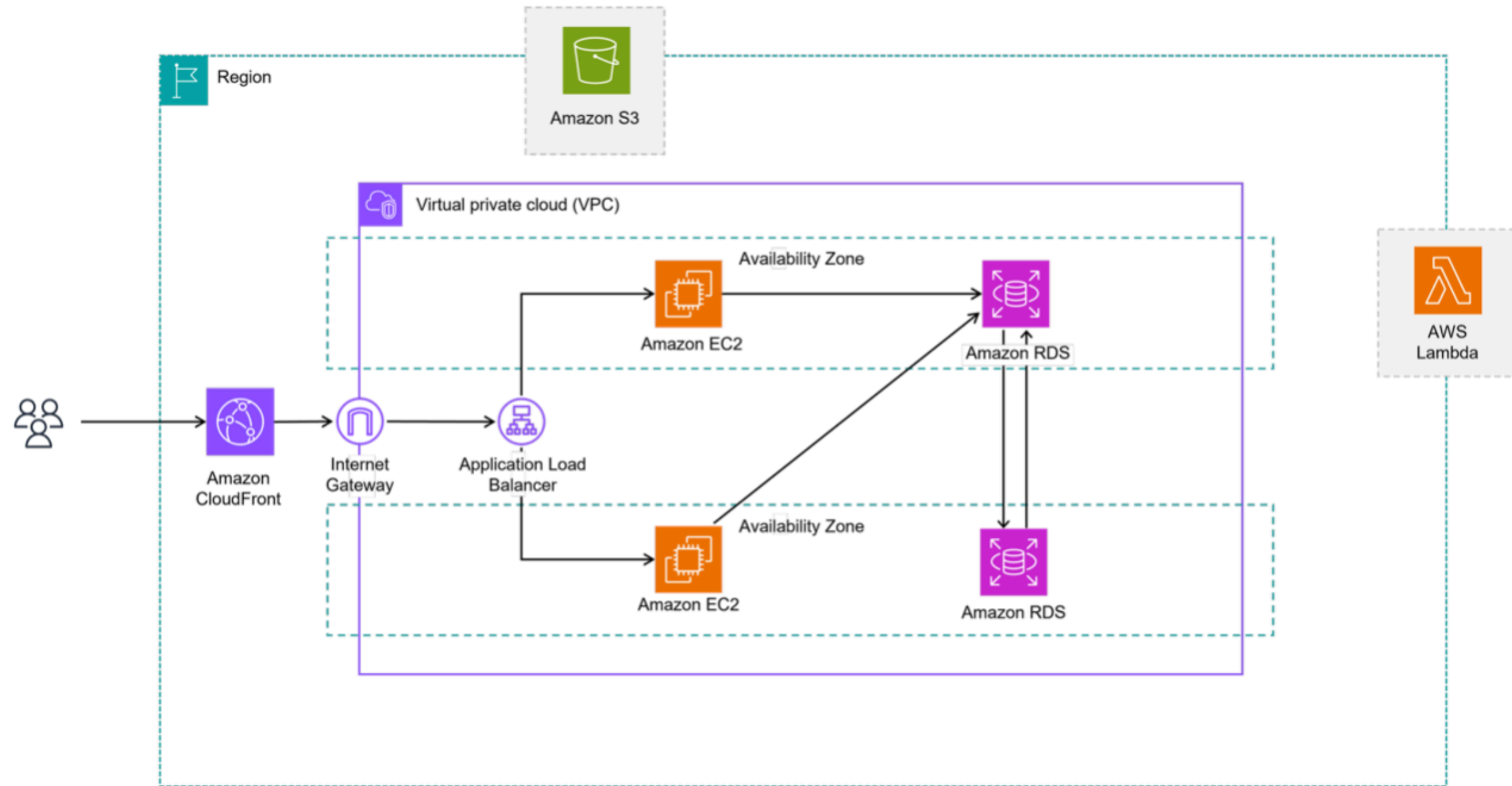
AWS Savings Plans
AWS Compute Optimizer
Amazon EC2 Spot Instances

ELB



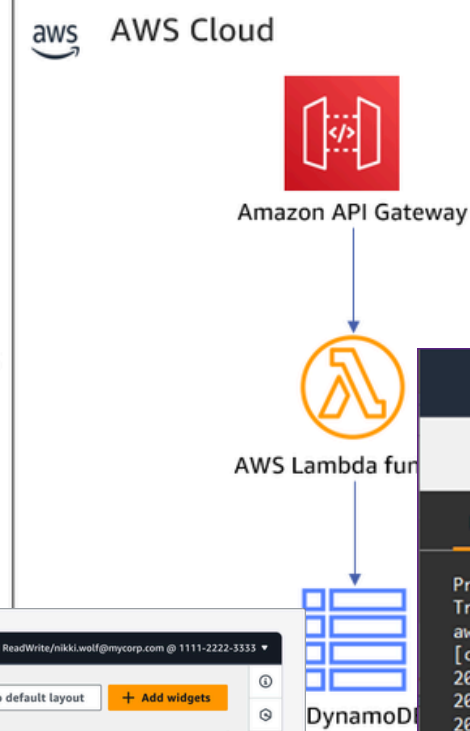
Elastic Load Balancing

Application Load Balancer
Network Load Balancer
Gateway Load Balancer



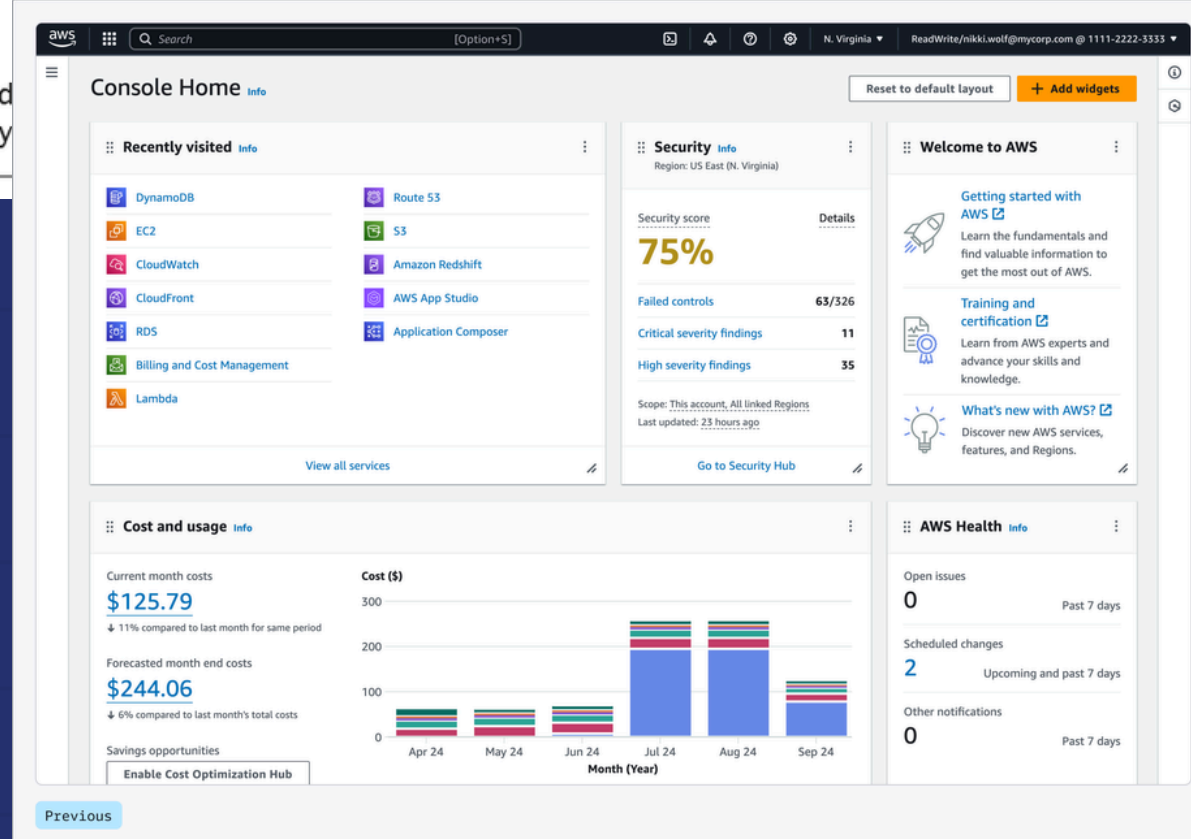
```
AWSTemplateFormatVersion: '2010-09-09'  
Transform: AWS::Serverless-2016-10-31  
Resources:  
  GetProductsFunction:  
    Type: AWS::Serverless::Function  
    Properties:  
      Handler: index.getProducts  
      Runtime: nodejs12.x  
      CodeUri: src/  
      Policies:  
        - DynamoDBReadPolicy:  
            TableName: !Ref ProductTable  
    Events:  
      GetResource:  
        Type: Api  
        Properties:
```

} creates



SDK

Prod
Ty



```
aws Services  
AWS CloudShell  
us-east-2  
Preparing your terminal...  
Try these commands to get started:  
aws help or aws command --help or aws command --cli-auto-prompt  
[cloudshell-user@ip-10-0-177-142 ~]$ aws s3 ls  
2020-08-06 20:53:18 amazon-braket-  
2020-06-30 09:38:31 appstream2-36fb080bb8-us-east-1-  
2020-06-30 11:42:41 audio-for-wordpress-74052805399a38f676f6a19c82e1d1e9a3913d5c  
2020-06-25 22:33:34 aws-blog-posts-content-usw  
2020-06-30 16:00:04 aws-deepracer-46116e97-6b45-4411-aead-0c7f2d8ef390  
2020-11-21 00:49:05 aws-panorama-jbarr  
2020-06-30 19:17:55 awsquest  
2020-06-30 19:18:23 awsroadtrip-backups  
2020-06-30 19:18:23 awsroadtrip-log  
2020-06-30 19:18:23 awsroadtrip-videos-raw  
2020-06-30 19:18:23 awsroadtrip.com  
2020-11-21 00:41:02 cf-templates-gpg2y3wgewtk-us-east-1  
2020-04-01 00:13:49 cw-syn-results-  
2020-12-05 22:03:58 do-not-delete-gatedgarden-audit-  
2020-07-03 13:57:51 jbarr  
2020-07-03 13:57:51 jbarr-audio-in  
2020-07-03 13:57:51 jbarr-audio-out  
2020-07-03 13:57:51 jbarr-aws-prod-logging  
2020-07-03 13:57:51 jbarr-awsquest  
2020-06-27 14:02:46 jbarr-backups  
2020-07-03 13:57:51 jbarr-batch-reports  
2020-07-03 13:57:51 jbarr-billing
```

CLI

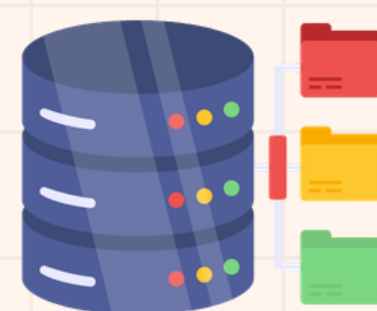
CONSOLE

AWS APPLIED USE CASES

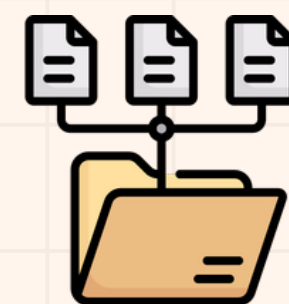
Scale
simulations
without
depending on a
local computer.

10.000.000 iterations

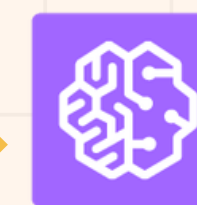
- Memory
- Computing time
- Storage



Amazon
EC2



amazon
ATHENA



Amazon SageMaker

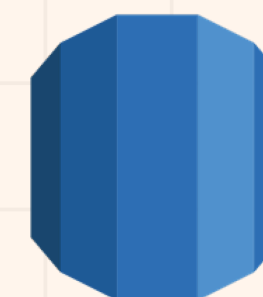
Architecture

Thousands of users

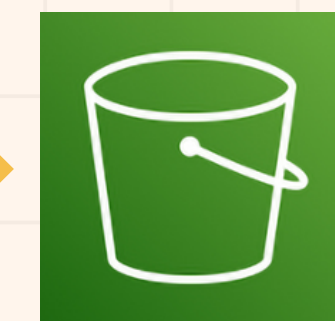
- Scaling
- Availability
- Storage
- Security



Amazon
EC2



amazon
RDS

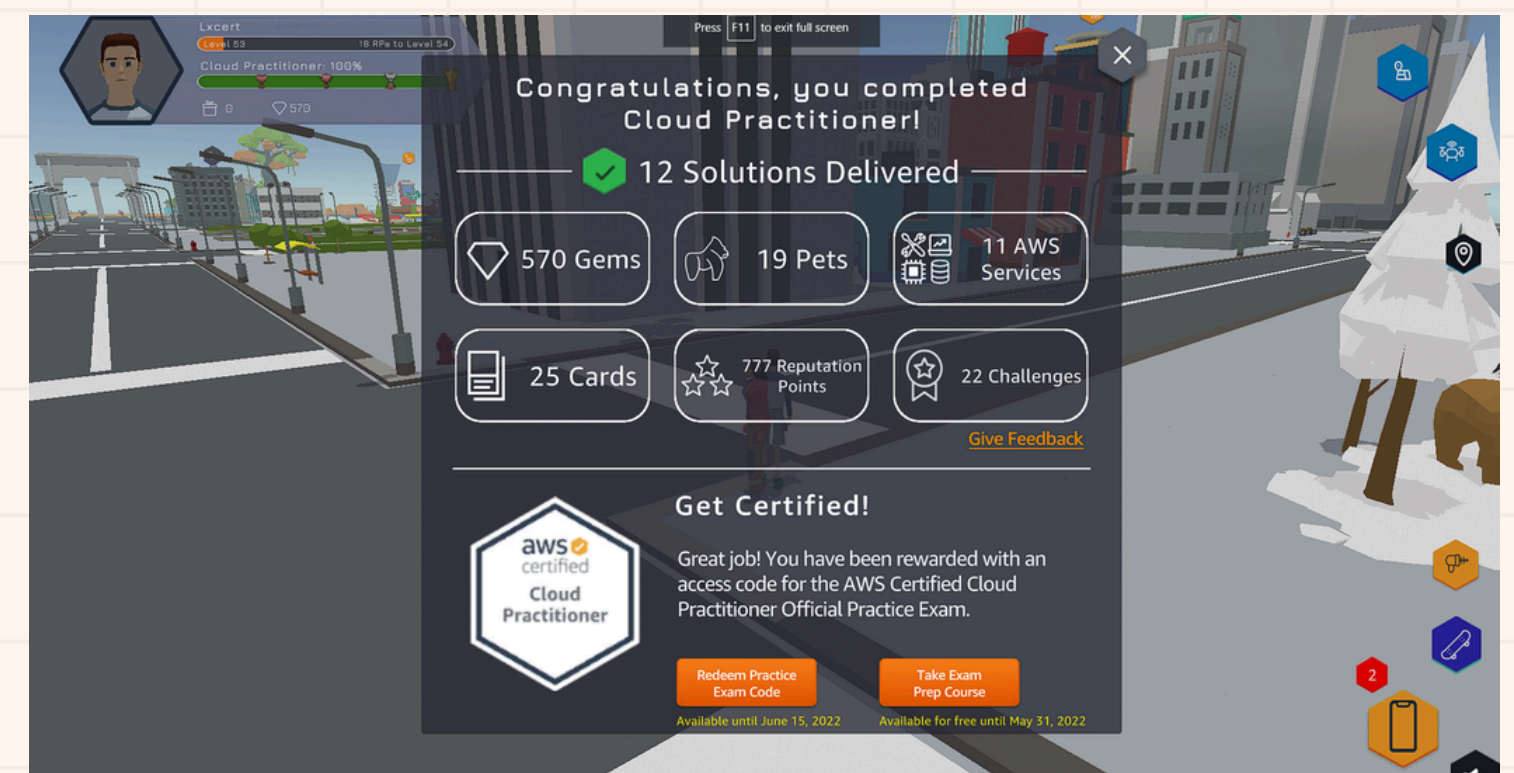


AWS APPLIED USE CASES



<https://skillbuilder.aws/learn>

<https://aws.amazon.com/marketplace>



<https://aws.amazon.com/es/training/digital/aws-cloud-quest/>

<https://www.youtube.com/watch?v=OXKgyQ8yCas>



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